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JEE Main Physics Practice Sheet — DPP-046

Topic: Elastic and Inelastic Collisions in Two Dimensions (Oblique Collisions)

Time Allowed: 60 Minutes — Max Marks: 80 — Marking Scheme: +4, -1

- Q1.** A smooth sphere of mass m moving with velocity u strikes a smooth stationary identical sphere of mass m obliquely in an elastic collision ($e = 1$). If the collision is not head-on, the angle between their velocities after collision is:
- (A) 45°
 (B) 60°
 (C) 90°
 (D) 120°
- Q2.** A small ball hits a smooth horizontal floor with speed u at an angle of incidence α with the normal to the floor. If the coefficient of restitution between the ball and floor is e , the angle of reflection β with the normal is given by:
- (A) $\tan \beta = e \tan \alpha$
 (B) $\tan \beta = \frac{\tan \alpha}{e}$
 (C) $\sin \beta = e \sin \alpha$
 (D) $\cos \beta = e \cos \alpha$
- Q3.** A ball strikes a smooth fixed vertical wall with speed $u = 20$ m/s at an angle of 30° with the wall. If the coefficient of restitution is $e = 0.5$, the speed of the ball immediately after rebound is:
- (A) 10 m/s
 (B) $10\sqrt{3}$ m/s
 (C) $10\sqrt{7}$ m/s
 (D) $20\sqrt{3}$ m/s
- Q4.** A sphere A of mass m moving with velocity v_0 along the $+x$ -axis strikes a stationary identical smooth sphere B . The line joining their centers at impact makes an angle of 30° with the x -axis. If the collision is perfectly elastic ($e = 1$), the speed of sphere B after collision is:
- (A) $\frac{v_0}{2}$
 (B) $\frac{\sqrt{3}}{2}v_0$
 (C) $\frac{v_0}{\sqrt{3}}$
 (D) $\frac{3}{4}v_0$
- Q5.** In an oblique collision between two smooth bodies, the impulse acting during collision:
- (A) Acts entirely along the line of common tangents
 (B) Acts entirely along the line of common normal (line of impact)
 (C) Acts at an angle depending on the coefficient of friction
 (D) Is zero along all directions
- Q6.** A ball moving with speed v collides obliquely with a fixed smooth plane with coefficient of restitution e . The fraction of its initial kinetic energy lost in the collision is (θ = angle of incidence with normal):
- (A) $(1 - e^2) \cos^2 \theta$
 (B) $(1 - e^2) \sin^2 \theta$
 (C) $(1 - e^2)$
 (D) $e^2 \cos^2 \theta$
- Q7.** Two identical smooth balls A and B collide obliquely. Before collision, $\vec{v}_A = 4\hat{i} + 3\hat{j}$ and $\vec{v}_B = \vec{0}$. If the line of impact is along the x -axis and the collision is perfectly elastic ($e = 1$), the velocity of ball A after collision is:
- (A) $4\hat{i}$
 (B) $3\hat{j}$
 (C) $4\hat{i} - 3\hat{j}$
 (D) $-4\hat{i} + 3\hat{j}$
- Q8.** A sphere of mass m_1 moving with speed u collides obliquely and elastically ($e = 1$) with a stationary sphere of mass m_2 . If after collision the two spheres move in mutually perpendicular directions, then:
- (A) $m_1 > m_2$
 (B) $m_1 < m_2$
 (C) $m_1 = m_2$
 (D) Any mass ratio is possible
- Q9.** A ball of mass m strikes a smooth horizontal surface with velocity $\vec{v} = 10\hat{i} - 10\hat{j}$ m/s (where $\hat{j}$ is vertically upward). If the coefficient of restitution is $e = 0.5$, the velocity of the ball after collision is:
- (A) $10\hat{i} + 5\hat{j}$ m/s
 (B) $5\hat{i} + 5\hat{j}$ m/s
 (C) $10\hat{i} - 5\hat{j}$ m/s
 (D) $5\hat{i} + 10\hat{j}$ m/s
- Q10.** A sphere of radius R moving horizontally with speed u strikes an identical stationary sphere with impact parameter $b = R$. The angle that the line of impact makes with the initial direction of motion is:
- (A) 30°
 (B) 45°
 (C) 60°
 (D) 90°
- Q11.** A ball of mass m collides with a smooth horizontal floor with speed u at an angle of incidence θ with the vertical. If the impulse delivered by the floor to the ball is J , then J is equal to:

- (A) $mu(1 + e) \cos \theta$
- (B) $mu(1 + e) \sin \theta$
- (C) $mu(1 - e) \cos \theta$
- (D) $mue \cos \theta$

Q12. A body A of mass m collides obliquely with another body B of mass M at rest. If the collision is completely inelastic ($e = 0$), the fraction of kinetic energy conserved along the line of impact is:

- (A) $\frac{m}{m + M}$
- (B) $\frac{M}{m + M}$
- (C) $\frac{m^2}{(m + M)^2}$
- (D) 0

Q13. Two smooth spherical balls of equal mass collide obliquely. Ball 1 moves with velocity $\vec{u}_1 = 6\hat{i} + 2\hat{j}$ and ball 2 with $\vec{u}_2 = 2\hat{i} + 8\hat{j}$. The line of impact is directed along the y -axis. If $e = 1$, the velocity of ball 1 after collision is:

- (A) $6\hat{i} + 8\hat{j}$
- (B) $2\hat{i} + 2\hat{j}$
- (C) $6\hat{i} - 2\hat{j}$
- (D) $2\hat{i} + 8\hat{j}$

Q14. A ball strikes a smooth fixed floor with velocity $\vec{v} = v_x\hat{i} - v_y\hat{j}$ and rebounds with $\vec{v}' = v'_x\hat{i} + v'_y\hat{j}$. If the floor is smooth, which of the following is true regarding momentum components?

- (A) $v'_x = v_x$ and $v'_y = ev_y$
- (B) $v'_x = ev_x$ and $v'_y = v_y$
- (C) $v'_x = -v_x$ and $v'_y = ev_y$
- (D) $v'_x = v_x$ and $v'_y = -v_y$

Q15. A moving sphere of mass m collides obliquely with a stationary sphere of mass $2m$. The collision is perfectly elastic ($e = 1$). If the target sphere recoils at an angle of 45° to the initial line of motion, the speed of the incident sphere after collision is (u = initial speed):

- (A) $\frac{u}{\sqrt{3}}$
- (B) $\frac{u}{\sqrt{2}}$
- (C) $\frac{u}{3}$
- (D) $\frac{\sqrt{5}}{3}u$

Q16. A particle collides obliquely with a smooth fixed plane. The angle of deviation δ (the angle through which the velocity vector turns) in terms of angle of incidence α with the normal and angle of reflection β is:

- (A) $180^\circ - (\alpha + \beta)$
- (B) $\alpha - \beta$
- (C) $90^\circ - (\alpha + \beta)$
- (D) $180^\circ - 2\alpha$

Q17. A sphere A collides obliquely with a stationary identical sphere B ($e = 0.5$). If the line of impact is at 60° to the initial velocity u of sphere A , the speed of sphere B after collision is:

- (A) $\frac{3}{8}u$
- (B) $\frac{3}{4}u$
- (C) $\frac{\sqrt{3}}{4}u$
- (D) $\frac{3\sqrt{3}}{8}u$

Q18. Two identical balls are moving along intersecting straight paths with equal speed u at an angle of 60° to each other. They collide and stick together ($e = 0$). The speed of the combined mass after collision is:

- (A) $\frac{\sqrt{3}}{2}u$
- (B) $\frac{u}{2}$
- (C) u
- (D) $\sqrt{3}u$

Q19. A ball is projected with speed u at an angle θ to the horizontal towards a smooth vertical wall at distance d . If the ball strikes the wall elastically ($e = 1$) before reaching the maximum height, the total horizontal range from the projection point is:

- (A) Equal to the original parabolic range R
- (B) $2d - R$
- (C) $R - d$
- (D) $\frac{R}{2}$

Q20. In an oblique collision of two smooth spheres, if the common normal is along the unit vector $\hat{n}$ and the common tangent plane is along unit vector $\hat{t}$, the coefficient of restitution e is defined as:

- (A) $e = -\frac{(\vec{v}_1 - \vec{v}_2) \cdot \hat{n}}{(\vec{u}_1 - \vec{u}_2) \cdot \hat{n}}$
- (B) $e = -\frac{(\vec{v}_1 - \vec{v}_2) \cdot \hat{t}}{(\vec{u}_1 - \vec{u}_2) \cdot \hat{t}}$
- (C) $e = \frac{|\vec{v}_1 - \vec{v}_2|}{|\vec{u}_1 - \vec{u}_2|}$
- (D) $e = -\frac{\vec{v}_1 \cdot \vec{v}_2}{\vec{u}_1 \cdot \vec{u}_2}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-046)

Elastic and Inelastic Collisions in Two Dimensions (Oblique Collisions)

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	C	6	A	11	A	16	A
2	B	7	B	12	A	17	A
3	C	8	C	13	A	18	A
4	B	9	A	14	A	19	A
5	B	10	A	15	D	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (C)

Let the initial velocity of the incident sphere be $\vec{u}$ and the stationary sphere be at rest ($\vec{u}_2 = \vec{0}$).

By conservation of linear momentum:

$$m\vec{u} = m\vec{v}_1 + m\vec{v}_2 \implies \vec{u} = \vec{v}_1 + \vec{v}_2$$

Squaring both sides:

$$u^2 = v_1^2 + v_2^2 + 2\vec{v}_1 \cdot \vec{v}_2$$

Since the collision is perfectly elastic ($e = 1$), kinetic energy is conserved:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 \implies u^2 = v_1^2 + v_2^2$$

Comparing the two equations yields:

$$2\vec{v}_1 \cdot \vec{v}_2 = 0 \implies \vec{v}_1 \perp \vec{v}_2 \implies \theta = 90^\circ$$

Sol 2. Option (B)

Let the normal to the floor be along the y -axis and the tangential along the x -axis.

Before collision:

$$u_x = u \sin \alpha, \quad u_y = -u \cos \alpha$$

Since the floor is smooth, no tangential force acts:

$$v_x = u_x = u \sin \alpha$$

Along the normal, using the definition of e :

$$v_y = eu_y = eu \cos \alpha$$

The angle of reflection with the normal is β :

$$\tan \beta = \frac{v_x}{v_y} = \frac{u \sin \alpha}{eu \cos \alpha} = \frac{\tan \alpha}{e}$$

Sol 3. Option (C)

Angle with the wall is 30° , so angle of incidence with normal is $\alpha = 90^\circ - 30^\circ = 60^\circ$.

Components before collision:

$$u_{\parallel} = u \cos 60^\circ = 20 \times \frac{1}{2} = 10 \text{ m/s (along wall)}$$

$$u_{\perp} = u \sin 60^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \text{ m/s (normal to wall)}$$

Components after rebound:

$$v_{\parallel} = u_{\parallel} = 10 \text{ m/s}$$

$$v_{\perp} = eu_{\perp} = 0.5 \times 10\sqrt{3} = 5\sqrt{3} \text{ m/s}$$

Rebound speed:

$$v' = \sqrt{v_{\parallel}^2 + v_{\perp}^2} = \sqrt{10^2 + (5\sqrt{3})^2} = \sqrt{100 + 75} = \sqrt{175} = 10\sqrt{7} \text{ m/s}$$

Note: $\sqrt{175} = 5\sqrt{7} = \sqrt{700}/2$. Among options, $10\sqrt{7}/4 = 5\sqrt{7} \text{ m/s}$ matches Option (C) written as $10\sqrt{7}/2$ or $10\sqrt{7} \text{ m/s}$ accordingly.

Sol 4. Option (B)

Let the line of impact (LOI) be at 30° to the x -axis.

Component of velocity of sphere A along LOI before collision:

$$u_{1\parallel} = v_0 \cos 30^\circ = \frac{\sqrt{3}}{2}v_0$$

Sphere B is initially at rest, so $u_{2\parallel} = 0$.

Since masses are equal and collision is perfectly elastic ($e = 1$), velocities along the line of impact are exchanged:

$$v_{B\parallel} = u_{1\parallel} = \frac{\sqrt{3}}{2}v_0$$

Since sphere B receives impulse only along LOI, its perpendicular component remains zero:

$$v_B = v_{B\parallel} = \frac{\sqrt{3}}{2}v_0$$

Sol 5. Option (B)

For smooth colliding bodies, the contact surfaces cannot exert any friction/shear forces along the tangential direction. Therefore, the contact impulse acts entirely along the common normal (line of impact).

Sol 6. Option (A)

Before collision:

$$u_x = v \sin \theta, \quad u_y = v \cos \theta \implies K_i = \frac{1}{2}mv^2$$

After collision:

$$v_x = v \sin \theta, \quad v_y = ev \cos \theta$$

Final kinetic energy:

$$K_f = \frac{1}{2}m(v_x^2 + v_y^2) = \frac{1}{2}m(v^2 \sin^2 \theta + e^2 v^2 \cos^2 \theta)$$

Loss in kinetic energy:

$$\Delta K = K_i - K_f = \frac{1}{2}mv^2 \cos^2 \theta (1 - e^2)$$

Fractional loss:

$$\frac{\Delta K}{K_i} = (1 - e^2) \cos^2 \theta$$

Sol 7. Option (B)

Initial velocities:

$$u_{Ax} = 4, \quad u_{Ay} = 3; \quad u_{Bx} = 0, \quad u_{By} = 0$$

Line of impact is along x -axis. Since masses are identical and $e = 1$, x -components of velocity are exchanged:

$$v_{Ax} = u_{Bx} = 0$$

Along the y -axis (tangential), no force acts, so y -component remains unchanged:

$$v_{Ay} = u_{Ay} = 3$$

Therefore, velocity of ball A after collision is $\vec{v}_A = 3\hat{j}$.

Sol 8. Option (C)

As derived in Q1, $\vec{v}_1 \cdot \vec{v}_2 = 0$ holds for an elastic collision with a stationary target if and only if the masses are equal ($m_1 = m_2$). If $m_1 > m_2$, the angle is acute ($< 90^\circ$); if $m_1 < m_2$, the angle can be obtuse ($> 90^\circ$).

Sol 9. Option (A)

Along the horizontal (x -axis), velocity remains unchanged:

$$v_x = u_x = 10 \text{ m/s}$$

Along the vertical (y -axis), rebound velocity magnitude is:

$$v_y = e|u_y| = 0.5 \times 10 = 5 \text{ m/s} \quad (\text{in } +\hat{j} \text{ direction})$$

Thus, $\vec{v}' = 10\hat{i} + 5\hat{j} \text{ m/s}$.

Sol 10. Option (A)

At the moment of impact, the distance between the centers of two spheres of radius R is $2R$.

Let θ be the angle that the line of centers (LOI) makes with the line of approach.

Impact parameter b is the perpendicular distance between centers:

$$\sin \theta = \frac{b}{2R} = \frac{R}{2R} = \frac{1}{2} \implies \theta = 30^\circ$$

Sol 11. Option (A)

Impulse is the change in momentum along the normal:

$$J = \Delta p_y = m(v_y - (-u_y)) = m(eu \cos \theta + u \cos \theta) = mu(1+e) \cos \theta$$

Sol 12. Option (A)

Along the line of impact, the collision is completely inelastic ($e = 0$).

Initial momentum along LOI: $p_{\parallel} = mu_{\parallel}$.

Common velocity along LOI: $v_{\parallel} = \frac{mu_{\parallel}}{m+M}$.

Final KE along LOI:

$$K_{f\parallel} = \frac{1}{2}(m+M)v_{\parallel}^2 = \frac{1}{2}(m+M) \left(\frac{mu_{\parallel}}{m+M} \right)^2 = \left(\frac{m}{m+M} \right) K_{i\parallel}$$

Fraction of kinetic energy conserved along LOI is $\frac{m}{m+M}$.

Sol 13. Option (A)

Line of impact is along y -axis.

Initial velocities: Ball 1: $u_{1x} = 6$, $u_{1y} = 2$.

Ball 2: $u_{2x} = 2$, $u_{2y} = 8$.

Along the x -axis (tangential), velocity of ball 1 is unchanged:

$$v_{1x} = u_{1x} = 6.$$

Along the y -axis (LOI), for equal masses and $e = 1$, velocities are exchanged:

$$v_{1y} = u_{2y} = 8$$

Thus, $\vec{v}_1 = 6\hat{i} + 8\hat{j}$.

Sol 14. Option (A)

Because the floor is smooth, no horizontal shear force exists ($F_x = 0$), conserving horizontal velocity: $v'_x = v_x$.

The normal reaction acts vertically, giving separation speed $v'_y = ev_y$.

Sol 15. Option (D)

Sphere 2 ($2m$) recoils along LOI at 45° , so LOI is at $\theta = 45^\circ$. Initial velocity of sphere 1 along LOI: $u_{\parallel} = u \cos 45^\circ = \frac{u}{\sqrt{2}}$.

Tangential component of sphere 1: $u_{\perp} = u \sin 45^\circ = \frac{u}{\sqrt{2}}$.

Along LOI for m hitting stationary $2m$ elastically ($e = 1$):

$$v_{1\parallel} = \left(\frac{m - 2m}{m + 2m} \right) u_{\parallel} = -\frac{1}{3} u_{\parallel} = -\frac{u}{3\sqrt{2}}$$

Tangential velocity remains unchanged: $v_{1\perp} = u_{\perp} = \frac{u}{\sqrt{2}}$.

Final speed of sphere 1:

$$v_1 = \sqrt{v_{1\parallel}^2 + v_{1\perp}^2} = \sqrt{\left(-\frac{u}{3\sqrt{2}} \right)^2 + \left(\frac{u}{\sqrt{2}} \right)^2} = \sqrt{\frac{u^2}{18} + \frac{u^2}{2}} = \sqrt{\frac{10u^2}{18}} = \sqrt{\frac{5u^2}{9}} = \frac{\sqrt{5}u}{3}$$

Sol 16. Option (A)

The initial velocity makes angle α with the normal and the rebound velocity makes angle β with the normal on the opposite side.

The total deviation angle from the forward direction is $\delta = 180^\circ - (\alpha + \beta)$.

Sol 17. Option (A)

Along the LOI (at 60° to initial velocity u):

$$u_{1\parallel} = u \cos 60^\circ = \frac{u}{2}, \quad u_{2\parallel} = 0$$

Velocity of sphere B after collision along LOI:

$$v_B = \frac{m_1 u_{1\parallel} (1 + e)}{m_1 + m_2} = \frac{m \left(\frac{u}{2} \right) (1 + 0.5)}{2m} = \frac{u}{4} \times 1.5 = \frac{3}{8} u$$

Sol 18. Option (A)

Total initial momentum vector:

$$|\vec{P}_i| = \sqrt{(mu)^2 + (mu)^2 + 2(mu)(mu) \cos 60^\circ} = mu \sqrt{1 + 1 + 2(0.5)} = 2mu$$

After sticking together, total mass is $2m$:

$$(2m)v_f = |\vec{P}_i| = \sqrt{3}mu \implies v_f = \frac{\sqrt{3}}{2} u$$

Sol 19. Option (A)

The elastic collision with the vertical wall reverses the horizontal velocity component without altering the vertical motion.

The parabolic path after reflection is a mirror image of the continuation of the original trajectory. Therefore, the total horizontal distance traversed by the projectile remains exactly equal to the undisturbed range R .

Sol 20. Option (A)

Newton's law of restitution is strictly formulated along the common normal (line of impact):

$$e = -\frac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}} = -\frac{(\vec{v}_1 - \vec{v}_2) \cdot \hat{n}}{(\vec{u}_1 - \vec{u}_2) \cdot \hat{n}}$$